FNLP
Lanuage Modelling

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Outline

1. **What are Language Models?**
   - Why do we need language models?
2. **N-gram Language Models**
   - Markov Assumptions
   - N-gram Language Models
   - Parameter Estimations
   - Evaluation
3. **Deal with Unseen Events**
   - Linear Interpolation
   - Smoothing
4. **Summary**

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  - the dog laughs
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  - syntax: grammars, syntactic analysis?
  - pragmatics?...

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  - pragmatics?...
  - **how about relying on data solely?**

Formally (sort of)

- Given a finite size of vocabulary:

$$\mathcal{V} = \{\text{a, an, dog, cat, bite, bark, ... zoo, ...}\}$$

- we can have an infinite set of strings of words, $\mathcal{S}$:
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  - the dog barks . $\rightarrow$ good
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  - ...

From a Probabilistic View

There is a probability distribution p over $\mathcal{S}$:

$$\sum_{s \in \mathcal{S}} p(s) = 1, p(s) \geq 0, \text{for all } s \in \mathcal{S}$$

that is :

- p(the dog barks .)
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- p(the dog laughs .) $\rightarrow$ large
- p(dog the bark a laugh smile .) $\rightarrow$ very small
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p: possibly what we want!

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- Optical Character Recognition (OCR)
- Handwriting Recognition
- Word Segmentation (*whatdoesthismean*?)
- Machine Translation
- Language Generation
- Question Answering/Dialogue/Chat

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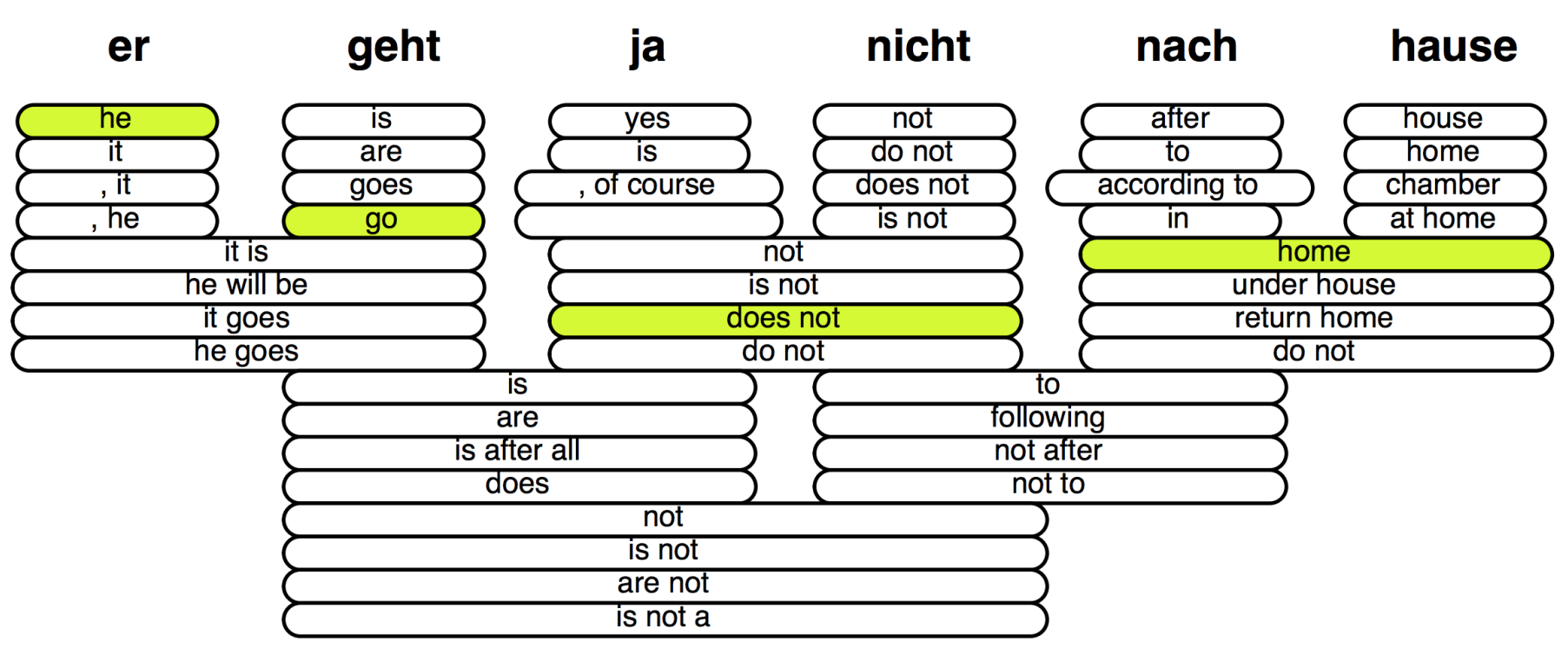
$$words^* = \arg\max_{words} \text{Faithfulness}(signals, words) + \text{Fluency}(words)$$

What can LM do?

Statistical Machine Translation



[from Philipp Koehn]

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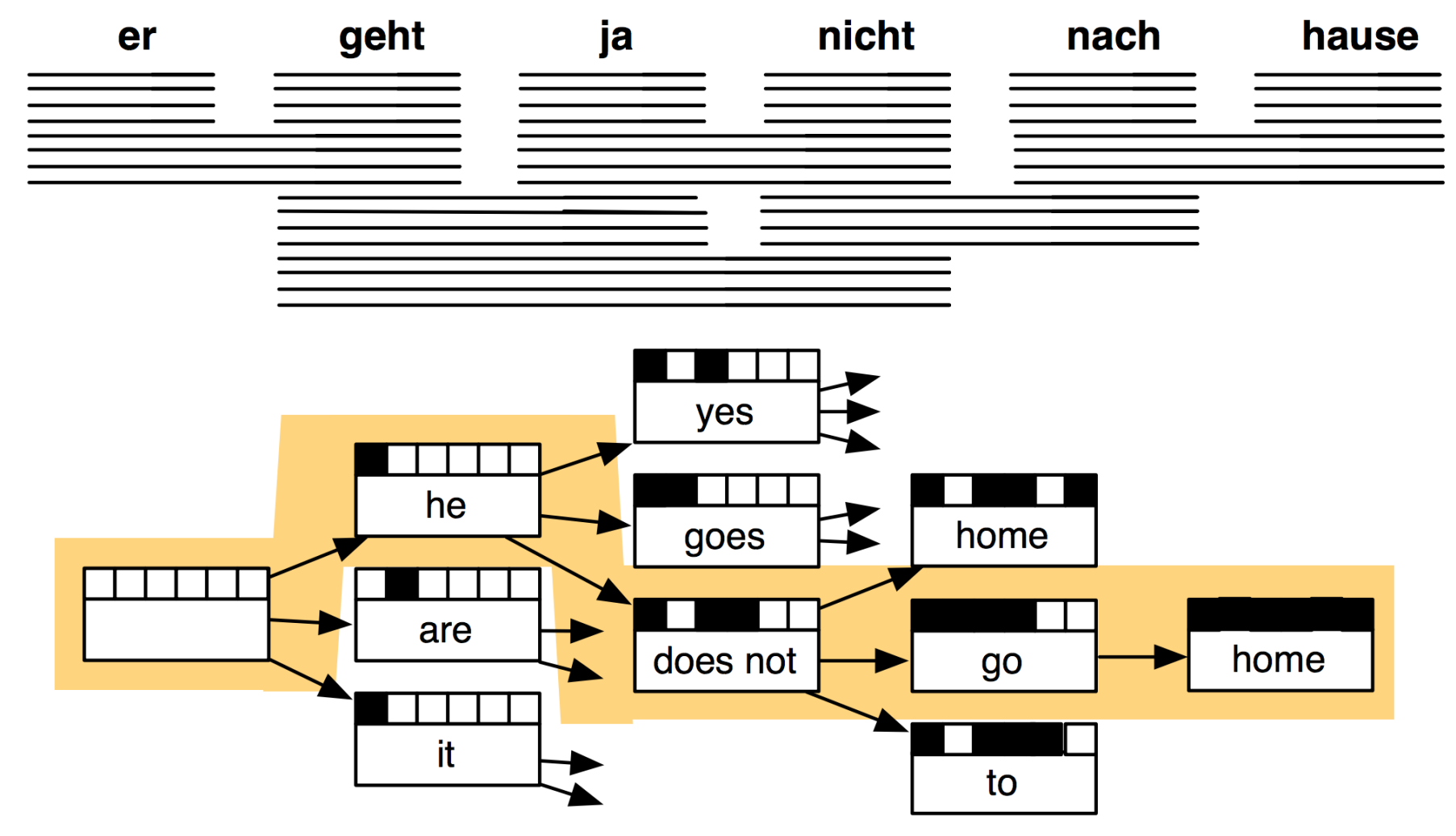

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What can LM do?

Text Generation

today the price of gold per ton , while production of shoe lasts and shoe industry , the bank intervened just after it considered and rejected an imf demand to rebuild depleted european stocks , sept 30 end primary 76 cts a share .

[from Chris Manning]

What can LM do?

Again, for all the above scenarios:

$$words^* = \arg\max_{words} \text{Faithfulness}(signals, words) + \text{Fluency}(words)$$

What can LM do?

Sentence Completion [MS SCC]

- That is his _ fault, but on the whole he's a good worker.
  - generous
  - mother's
  - successful
  - favorite
  - main
- Was she his _, his friend, or his mistress?
  - client
  - musings
  - discomfiture
  - choice
  - opportunity
- All red-headed men who are above the age of _ years, are eligible.
  - seven
  - twenty-one
  - 1,200
  - 60,000
  - 800

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- the so-called **surprisal theory** (Hale, 2001; Levy, 2008)
- *words are easier to comprehend in contexts where they are highly predictable than in unconstraining contexts*

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  - how big is the $\mathcal{S}$?
  - s can be in various forms
  - can we collect $count(s)$ as before?
  - can we properly estimate $p(s)$?
- Break down?

Another Point of View

- Sentence as a sequence of words $w_1, w_2, w_3, ..., w_n$
- Sentence as a sequence of random variables $X_1, X_2, X_3, ..., X_n$
- The model : $P(X_1 = w_1, X_2 = w_2, X_3 = w_3, ..., X_n = STOP)$
  - This is a **sequence**, not a set!
  - $P(X_1 = \text{I}, X_2 = \text{love})$ and $P(X_1 = \text{love}, X_2 = \text{I})$ are different
  - STOP and, sometimes, START

A bit Probabilistic

$$P(X_1 = w_1, X_2 = w_2, X_3 = w_3, ..., X_n = STOP)$$
$$= P(X_1 = w_1) \prod_{i=2}^{n} P(X_i = w_i | X_1 = w_1, ..., X_{i-1} = w_{i-1})$$

- Chain Rule

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- when n is big, ...

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- Let's make ASSUMPTIONS! $\Rightarrow$ First-order Markov assumption:

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Make Assumption

Let's make stronger ASSUMPTIONS! $\Rightarrow$ no history at all:

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- Unigram Language Model
- under MLE estimations:

$$P(X_i = w_i) = \frac{count(w_i)}{\sum_{w_*} count(w_*)}$$

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- This is actually bag-of-words (BoW), definitely not good enough

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Bigram and Trigram LMs

- Widely used in many applications!
- No history assumption → Unigram LM
- First-order Markov assumption → Bigram LM
- Second-order Markov assumption → Trigram LM
- 4-gram LM, 5-gram LM, ...

Trigram LM

- A finite vocabulary $\mathcal{V}$
- Parameters $p(a|b,c)$, where b, c, a is an arbitrary trigram $a \in \mathcal{V} \cup \{STOP\}$ and $b, c \in \mathcal{V}$
- For a new sentence $w_1, w_2, ...w_n$

$$p(w_1, w_2, ...w_n) = p(w_1)p(w_2|w_1)\prod_{i=3}^{n} p(w_i|w_{i-2}, w_{i-1})$$

A Toy Example

For the sentence:

the cat laughs . STOP

given a trigram LM, we have:

$$
\begin{aligned}
p(\text{the cat laughs . STOP}) = & p(\text{the}) \\
& \times p(\text{cat}|\text{the}) \\
& \times p(\text{laughs}|\text{the, cat}) \\
& \times p(.|\text{cat, laughs}) \\
& \times p(\text{STOP}|\text{laughs, .})
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- Maximum likelihood estimation
- More data lead to better model?

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- 12,537,755 v.s. 51,505,109,010,436 ($\sim$ 1/4,000,000)
- Data sparsity: most are zeros
- Trigram LMs are often good enough

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- the lower perplexity, the better LM

Perplexity

For a dataset $\mathcal{D}$, with vocabulary $\mathcal{V}$,

- if for any trigram, we have $p(a|b,c) = \frac{1}{|\mathcal{V}|+1}$ (a uniform distribution over $\mathcal{V} \cup STOP$) for all $b, c \in \mathcal{V} \cup \{*\}$
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- Perplexity $=|\mathcal{V}| + 1$

- Perplexity is a measure of *branching factor*
- Considered as the *effective size of the vocabulary*
- Also, to evaluate how hard an NLP task is?

Evaluating Language Models

- use the `Training dataset` to estimate the n-grams, possibly tuned
- compute perplexity on the `Test dataset`

However,

- Better to use test sets that are widely accepted by the community.
- **Perplexity** scores should be compared against the same $\mathcal{V}$.
- Perplexity is NOT a universal metric for downstream applications.

Unigram v.s. Bigram v.s. Trigram ...

Intuitively, which one is better?

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- if k is smaller, ...
  - Does your model learn about the language?
- if k is bigger, ...
  - The cost!
  - The engineering thing! (zeros)
  - ...

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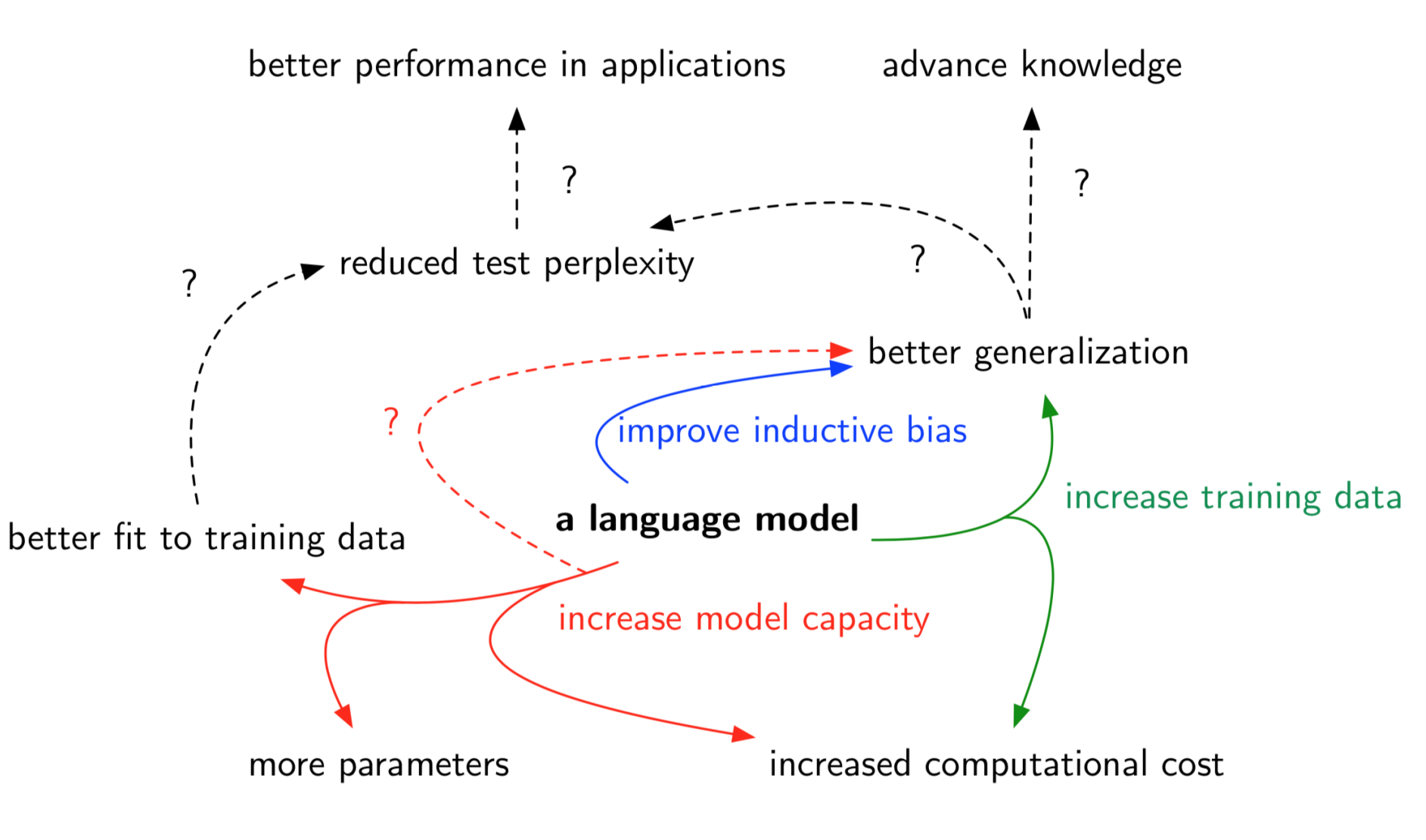
- if k is smaller, ...
  - Does your model learn about the language?
- if k is bigger, ...
  - The cost!
  - The engineering thing! (zeros)
  - ...
- if you have more training data, ...
  - perform better
  - generalize better
  - take a look at Google's N-gram

Language Modelling Research



[from Noah Smith]

An Example

Training set

- i love pku .
- i like thu .
- you love pku .
- you do not like thu .

- Can you estimate a bigram LM from the training data?
  - $p(\text{thu}|\text{like})$
  - $p(\text{like}|\text{i})$

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Test set

- you like pku .
- i hate thu .

- Can you compute the perplexity using the bigram LM ?
  - $p(\text{you, like, pku, ..., STOP})$
  - $p(\text{i, hate, thu, ..., STOP})$

Outline

1. What are Language Models?
   - Why do we need language models?
2. N-gram Language Models
   - Markov Assumptions
   - N-gram Language Models
   - Parameter Estimations
   - Evaluation
3. **Deal with Unseen Events**
   - Linear Interpolation
   - Smoothing
4. Summary

Unseen Events

- New words
  - *hate*
- New n-grams
  - *you like*, *like pku*

Unseen Events

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  - *hate*
- New n-grams
  - *you like*, *like pku*

Recall the MLE estimation ...

Unseen Events

MLE yields bad estimations for Unseen Events

- many Zeros

Unseen Events

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Brain Storm

- What can we do?

Unseen Events

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We should give some probability mass to those unseen events! This will inevitably reduce the mass for seen events.

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- **back-off** to a lower order estimations
- produce a pseudo word UNKNOWN
- smooth our raw estimations

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If we do not have a higher-order ngram, can we use less context, i.e., related lower-order ngrams ?

Linear Interpolation

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If we do not have a higher-order ngram, can we use less context, i.e., related lower-order ngrams ?

- Back-off
  - use higher-order estimators if you have enough counts
  - otherwise lower-order, ...
- Interpolation
  - just combine estimators of different orders:

$$\begin{aligned} p_{il}(w_i|w_{i-2}, w_{i-1}) = & \quad \lambda_1 p_{ml}(w_i|w_{i-2}, w_{i-1}) \\ + & \quad \lambda_2 p_{ml}(w_i|w_{i-1}) \\ + & \quad \lambda_3 p_{ml}(w_i) \end{aligned}$$

    where $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and $\lambda_i \geq 0$ for all i
  - does it still yield a distribution?

Stupid Back-off

- for a fixed vocabulary at a web scale, we have trigrams :

$$s(w_i|w_{i-2}, w_{i-1}) = \begin{cases} \frac{count(w_{i-2},w_{i-1},w_i)}{count(w_{i-2},w_{i-1})} & count(w_{i-2}, w_{i-1}, w_i) > 0 \\ 0.4 s(w_i|w_{i-1}) & \text{otherwise} \end{cases}$$

- Brants et al. 2007 (Google)

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- is this model a probabilistic one?
- NO! $s()$s do not sum to 1

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A Basic Smoothing Solution

Just add **one count** for all possible n-grams!

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Just add **one count** for all possible n-grams!

- Does this work ?
- Any side effect you could imagine ?

Add-One Smoothing

Intuition

For all possible n-grams, just add an extra count.

$$p_{add1}(w_i|w_{i-1}) = \frac{count(w_{i-1}, w_i) + 1}{count(w_{i-1}) + |\mathcal{V}|}$$

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Example

- $|\mathcal{V}| = 12$, $\mathcal{V} = \{\text{horse}, \text{cat}, \text{dog}, \text{dunkey}, \text{the}, ...\}$
- seen: $p(\text{horse}|\text{the}) = 0.5$, $p(\text{cat}|\text{the}) = 0.5$
- 0 for the unseen
- after Add-one
  - previously seen: $p_{add1}(\text{horse}|\text{the}) = 0.18$, $p_{add1}(\text{cat}|\text{the}) = 0.18$
  - previously unseen: $p_{add1}(\text{dog}|\text{the}) = 0.09$, $p_{add1}(\text{dunkey}|\text{the}) = 0.09$, ...

Add-One Smoothing

- $p(\text{SEEN EVENTS}|\text{the})$ from 1 to 0.36
- $p(\text{UNSEEN EVENTS}|the)$ from 0 to 0.64

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Add-One Smoothing

- $p(\text{SEEN EVENTS}|\text{the})$ from 1 to 0.36
- $p(\text{UNSEEN EVENTS}|the)$ from 0 to 0.64
- are we putting too much probability mass on unseen events?
- too many zeros in our example
- suit cases where $\#(\text{UNSEEN})$ is not that huge

Good-Turing Discounting

Intuition: Use the counts of n-grams that are **seen once** to help estimate the counts of **unseen events**

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- count of count : N_r, number of words with frequency r
- Translate real counts r into adjusted counts r^*:

$$r^* = (r+1)\frac{N_{r+1}}{N_r}$$

- The probability mass reserved for unseen events is N_1/N_{all}
- For larger r (where N_{r-1} is often 0), various other methods can be applied (curve fitting or linear regression).

Good-Turing Discounting

During testing, how possible we see a word that appears r times in the training set? Imagine we can randomly delete a word from the training set to construct a **new but pseudo** training set:

- everytime, we delete a word w
  - we know w appears $r+1$ times in the original set
  - now it is supposed to appear r times in the new training set
- the total apearances of $r+1$ words in the original set is $N_{r+1}(r+1)$
- after many times of deletion, the number of those words in the new set could be considered as N_r
- the possible/average counts of those words should be $N_{r+1}(r+1)/N_r$

Good-Turing Example

Trained on 22m words of AP News (Church and Gale, 1991)

r	r^*	$r - r^*$
0	0.000027	
1	0.446	0.554
2	1.26	0.74
3	2.24	0.76
4	3.24	0.76
5	4.22	0.78
6	5.19	0.81
7	6.21	0.79
8	7.24	0.76

Absolute Discounting

In Good-Turing discounting,

- It looks like $r^* = r - 0.75$, in most cases

So, we may

- Subtract a fixed number d from each count

$$p_{abd}(w_2|w_1) = \frac{count(w_1, w_2) - d}{count(w_1)} + \lambda(w_1)p(w_2)$$

- Typical counts 1 and 2 are treated differently
- $\lambda(*)$: a weighting function

Kneser-Ney Smoothing

- Popular in the traditional category
- Combines various smoothing method
  - absolute discounting
  - considers diversity of history, predicted words
  - interpolation

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- Really easy to build, on billions of words
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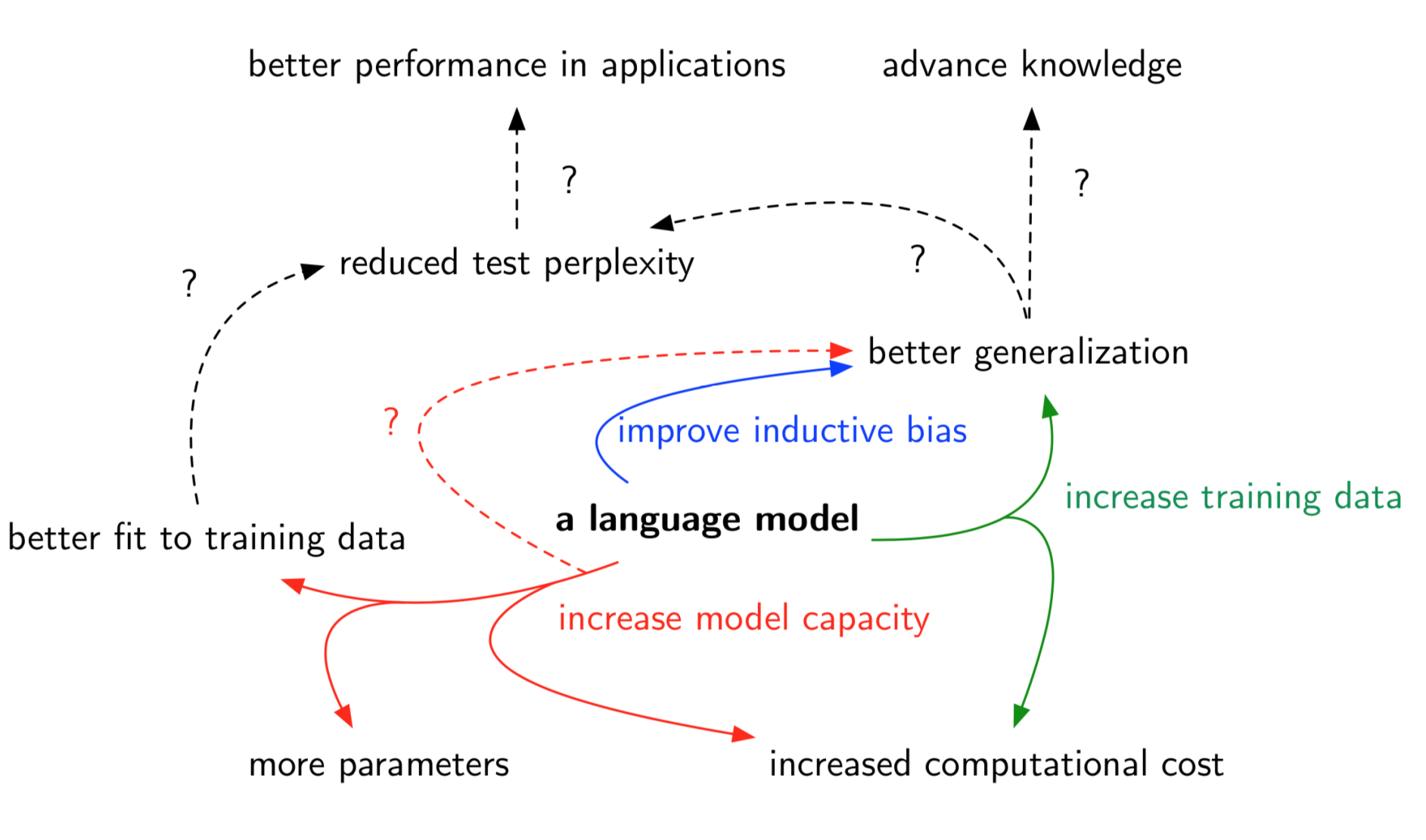

[from Noah Smith]

Summary

What you should know about N-gram LMs

- Why do we make Markov assumptions
- How to build bi-gram LMs, tri-gram LMs, ...
- Perplexity and language model evaluation
- Interpolation and smoothing/discounting

Now

Questions

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- can we go beyond *counting words*?
- can we use something else learned before?
- can we train on *much more words*?

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- can we go beyond *counting words*?
- can we use something else learned before? *classification*?
- can we train on *much more words*?

Readings

- Chapter 3, Speech and Language Processing (3rd. **SLP**)
- SRI Language Modeling Toolkit, http://www.speech.sri.com/projects/srilm/manpages/
- Improved backing-off for m-gram language modeling, Reinhard Kneser and Hermann Ney, ICASSP 1995.
- A comparison of the enhanced Good-Turing and deleted estimation methods for estimating probabilities of English bigrams, Kenneth W.Church and William A.Gale, Computer Speech & Language, 1991, V5-1, pp19-54
- Large Language Models in Machine Translation, Thorsten Brants, Ashok C. Popat, Peng Xu, Franz J. Och, Jeffrey Dean, EMNLP 2007
- Google N-grams: https://blog.research.google/2006/08/all-our-n-gram-are-belong-to-you.html
- Google Book N-grams: https://books.google.com/ngrams/

Quizz

Here, we collect a frequence table for a bi-gram model on a samll toy dataset. Each cell stores the occurrence of a pair of words, e.g., *saw* appears 30 times right after *I*.

w_1/w_2	I	finished	work	saw	the	beautiful	gift
I	0	40	20	30	0	0	0
finished	0	0	10	0	8	5	3
work	0	5	0	0	0	0	0
saw	0	5	5	0	8	5	3
the	0	10	15	0	0	20	10
beautiful	0	0	5	0	0	0	10
gift	0	5	0	0	0	3	0

Q1: What are the most likely four-word sentence beginning with I?

I _ _ _.

Q2: What is the probability that *gift* appears after *the* ?